DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #03C: MARTIAL ARTS & TECHNIQUES (PART 3 — MANTRAS, INNER QI CIRCULATION, MEDITATION HALOS, WEATHER RESONANCE & GRAND FORMATION WARFARE)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Series Context**:
  - `03A`: Active Combat Moves (`COMBAT_TECHNIQUE` / `moveDefs`), `SK.list` Caching, `chooseSkill` AI, 16 Weapon Mismatches & Spatial Range.
  - `03B`: Composable Keyword Engine, 34 Multi-Tier Elemental Reactions (`DARK_YIN` vs `YIN_SHADOW`), Passive Slots & Duets.
  - **`03C` (This Document)**: Dedicated, exhaustive deep dive into **Mantras & Inner Kungfu Scriptures (`MANTRA` / 心法 — 122 `JD.TECH.mantras` + 42 `ART190_*_MANTRA` + 31 Grand Dao Mantras)** across cultivation, lifespan, stat formulas, meditation auras, walking breathing, combat dharma procs, and autonomous NPC decision-making.

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never touch `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order**: Preserve exact load sequence in `index.html` and `test/load.js`.
3. **100% Emergent NPC Autonomy**: Players **cannot** micro-command NPCs. Every Mantra equip choice, Qi circulation stance (`SERENE_PURIFICATION` / `HARMONIOUS_CIRCULATION` / `OVERDRIVE_COMBUSTION`), meditation seat selection, and Sect Grand Formation stance must be decided autonomously by NPCs and Sect Officers based on `HC` traits, physiology, injuries, toxicity, and battlefield threats.
4. **Test-Lock Awareness**: Note that `test/v550_mantra_resonance_and_formation_warfare_suite.js:69` currently asserts `JD.TECH.mantras.length === 122`. Any catalog unification must either register Grand Dao mantras into a unified runtime index (`CULT.allMantras` / `CULT.mantraById`) while keeping raw `JD.TECH.mantras` backward-compatible OR

update `v550`'s hardcoded count assertion intentionally alongside full regression verification.

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1. `[MANTRA-DATA-01]` The Two-Table Mantra Schism, Missing `GOD` Tier-1 Starter, & 31-Path Stat Mismatch

1.1 Split Storage Between `JD.TECH.mantras` (122) and `JD.TECH.techniques` (42 `ART190` + 31 Grand Dao)

- **Locations**: `data/data_tech.js`, `data/data_martial.js`, `js/cultivation.js`, `test/v230_grand_dao_deep_parity_and_universal_arsenal_suite.js:82–83`, `test/v550_mantra_resonance_and_formation_warfare_suite.js:69–75`
- **Root Cause**:
  1. Because `v550` locked `assert.strictEqual(JD.TECH.mantras.length, 122)`, patches `v2.0.0–v2.3.0` could not append Mantras for the **31 newer Grand Dao paths (`paths 13..43`)** into `JD.TECH.mantras`. Instead, they placed `MANTRA_*` entries inside `JD.TECH.techniques`!
  2. Meanwhile, 42 `ART190_*` entries inside `JD.TECH.techniques` (`ART190_XIAN_002, 006, 010...`, `ART190_BODY_002, 006, 010...`, `ART190_GOD_002, 006, 010...`) carry `cat: 'QI_MANTRA' | 'BODY_MANTRA' | 'FAITH_MANTRA'` AND were enriched in `v550` with `mantraArchetype` (e.g., `ART190_XIAN_010` $\to$ `VAJRA_IMMORTAL_BODY_BELL`), yet they also have 3 named active combat moves (`moves: [...]`)!
- **Cascading Consequences**:
  - **Asymmetric Mantra Progression Ladder**: The original 12 paths have rich 8–14 Mantra ladders spanning Tiers 1–7 in `JD.TECH.mantras`, whereas the 31 newer Grand Dao paths have minimal Mantra variety in `JD.TECH.techniques`, causing high-stage Grand Dao elders to stay stuck on low-tier starter mantras unless universal mantras exist.
  - **`GOD` Path Has Zero Tier-1 Starter Mantra (`data_tech.js:551`)**:
    - `XIAN` has `MANTRA_BASIC_BREATH` (`tier: 1, power: 15`), `BODY` has `MANTRA_IRON_SKIN` (`tier: 1, power: 18`), `DEMON` has `MANTRA_DEMON_SHADOW_BREATH` (`tier: 1, power: 16`), `HEART_SOUL` has `ART_V112_SOUL_CLEAR_MIRROR_ZEN_T1` (`tier: 1, power: 16`).
    - **`GOD`'s lowest Mantra in `data_tech.js` is `MANTRA_INCENSE_PRAYER` at `tier: 2, power: 35`!** Either Stage-1 `GOD` initiates cannot learn a Tier-1 Mantra from the library (`reqTier: 2 > stage: 1`), or when

force-assigned on awakening they start with `2.3x` the Mantra power of every other path at Stage 1!

1.2 Wrong Combat Stat Scaling (`+matk`) Granted by Mantras to 15+ Physical & Zero-Qi Grand Dao Paths (`js/stats.js`)

- **Locations**: `js/stats.js` (`compute` & `getBaseFast` Mantra bonus block)
- **Root Cause**:
  - When computing base stats from `n.cult.mantra`, `stats.js` checks a short hardcoded list of early physical paths (`BODY`, `BEAST_SOVEREIGN`, `MECH_PUPPET`) to grant `+atk / +def / +hp`, and sends **every other path into a default `else` branch that grants `+matk` and `+mdef`**!
- **Absurd Runtime Impact**:
  1. **Physical Grand Dao Paths (`ARENA_GLORY`, `DEVOURING_FLESH`, `PRIMORDIAL_SHIFTER`, `NINPO_SHADOW`, `GRAVITY_DAO`, `WANDERING_DAO`)**: Upgrading from a Tier-1 Mantra (`power: 18`) to a Tier-7 Mantra (`power: 355`) grants **`+140+ Magic Attack (matk)` and `+0 Physical Attack (atk)`**, doing nothing for their physical claw/fist/blade/spear strikes!
  2. **Contextual & Lore Contradiction on `SHADOW_SAMURAI` (ไร้ตันเถียน/ไร้ปราณ — `qiMax === 0`)**:
     - By lore and design (`CULT.PATHS.SHADOW_SAMURAI`), Shadow Samurai have **zero Qi (`qiMax(n) === 0`)** and rely purely on physical blade mastery and `IAI_FOCUS` mental concentration (*Mindfulness / Bushido Heart-Method*, จิตสมาธิบูชิโด).
     - Yet `stats.js` gives their Mantra `+matk`, `MANTRA_SYSTEM` assigns them Qi-breathing archetypes that grant `+45% Qi regen`, `-30% Flight Qi cost`, or `+18% Qi siphon` (all multiplying `0`!), and UI labels describe `"การเดินลมปราณสู่ตันเถียน"`!
- **Remediation Blueprint**:
  - Drive Mantra stat allocation in **both** `ST.compute` and `ST.getBaseFast` from `CULT.PATHS[c.path].statWeights` (if `atk >= matk * 1.15`, grant `atk + def + hp`; if balanced hybrid, split `atk + matk`; if caster, grant `matk + mdef`).
  - For `SHADOW_SAMURAI`, treat `c.mantra` in lore/UI and mechanics as **Bushido Zanshin / Mind-Focus Method (**เคล็ดสมาธิจิตดาบไร้ปราณ**)**: convert any `qiRegen` / `qisteal` / `qiCostReduction` from `MANTRA_SYSTEM` into `+crit / +acc / +stamina recovery / +IAI_FOCUS gain` (`+3` gauge on hit) instead of dead `0`-Qi math!
  - Add a true `tier: 1, power: 16` starter `FAITH_MANTRA` for `GOD` (or normalize `MANTRA_INCENSE_PRAYER` to `tier: 1, power: 17` and add a `tier: 3` bridge `FAITH_MANTRA`).

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2. [MANTRA-CMB-02] Critical Combat Bugs in CULT.MANTRA_SYSTEM (combat.js:263–299)

Inspecting `js/combat.js:263–299` inside `COMBAT.applyDamage(W, f, a, d, dmg, note)` reveals **four severe live combat bugs**:

2.1 Bug 2A — Uninitialized `mantraState` Guard Silently Disables All Mantra Combat Dharma!

- **Code (`combat.js:265, 280`)**:

```JavaScript
if (a && !a.foe && a.cult && a.mantraState) { ... }
if (d && !d.foe && d.cult && d.mantraState) { ... }
```

- **Failure**: A newly awakened NPC, recruited wanderer, or defender who enters combat before `mantraState` was lazily created has `a.mantraState === undefined`, even though `a.cult.mantra` is valid! Both blocks are completely skipped (`0%` Mantra Dharma effect) until something else initializes `mantraState`.
- **Fix**: Check `a.cult && a.cult.mantra` and let `CULT.MANTRA_SYSTEM.resolveNpcMantra(W, a)` lazily ensure `a.mantraState` (defaulting to `'HARMONIOUS_CIRCULATION'`).

2.2 Bug 2B — `EVERGREEN_LONGEVITY_SPRING` Heals `3.5% MaxHP` Per `applyDamage` Call (Up to `+63% MaxHP` in One AoE Cast & Infinite Healing From DoT Ticks!)

- **Code (`combat.js:274–276`)**:

```JavaScript
} else if (amr.archetypeId === 'EVERGREEN_LONGEVITY_SPRING') {
  const regenHp = Math.max(1, Math.round(C.hpMax(a) * (amr.spec.combatRegenHpPct
|| 0.035)));
  C.setHp(a, Math.min(C.hpMax(a), C.hp(a) + regenHp));
```

- **Failure**:
  - `ACTIVE_PATCH_PROMPT_V5_5_0.md` specified: **"restores `3.5%` HP per combat round"**.

- Because this code was placed raw inside `C.applyDamage` with **no ICD (`_mantraAtkIcd`), no `healMultiplier` (`healBlock` / `stygianFreeze`) check, and no DoT-tick exclusion**:
    1. Casting a 3-hit AoE across 6 enemies calls `C.applyDamage` **18 times in the same tick**, instantly healing `18 × 3.5% = +63% MaxHP`!
    2. Every poison/bleed/burn DoT tick in `C.statusTick` (`combat.js:959`) calls `C.applyDamage(W, f, src, u, dotDmg)`, healing the poisoner `a` for `+3.5%` of `a`'s MaxHP per poisoned enemy every single tick—even while `a` is stunned or afflicted with `stygianFreeze` (`0%` healing)!
    3. `DEVOURING_NETHER_VORTEX` (`lines 270–273`) similarly siphons HP & Qi on every DoT tick and ignores `C.healMultiplier(a, W.t)`!

2.3 Bug 2C — `VAJRA_IMMORTAL_BODY_BELL`, `SWORD_QI_MERIDIAN`, & `PRIMORDIAL_YANG_FURNACE` Reflect Lasers Across 25 Tiles on Poison/Bleed Ticks!

- **Code (`combat.js:282–296`)**:
    - When `C.statusTick` ticks `poison`, `bleed`, or `burn` on defender `d`, it passes the poison's `src` as `a` (`note` is `undefined`).
    - `VAJRA_IMMORTAL_BODY_BELL` (`line 286`), `SWORD_QI_MERIDIAN` (`line 292`), and `PRIMORDIAL_YANG_FURNACE` (`line 295`) **do not check whether the damage came from a DoT/Domain tick, nor check melee distance (`unitDist(a, d) <= 2.4`), nor route through the unified `COMBAT.reflectOf` cap**!
    - **Absurd Result**: Every time poison ticks inside a Vajra monk's bloodstream, their "Golden Bell melee reflection" blasts the archer/alchemist standing 25 tiles away across the battlefield!

2.4 Bug 2D — `stMult` (`OVERDRIVE_COMBUSTION = 1.28x`) Applied Multiplicatively as Final Post-Armor Damage Only on 2 of 7 Archetypes, and `SWORD_QI_MERIDIAN` Ignores Actual Penetration & Weapon!

- **Code (`combat.js:267–278`)**:
    - `PRIMORDIAL_YANG_FURNACE`: `dmg = Math.round(dmg * 1.15 * 1.28)` $\to$ **`+47.2%` final damage multiplier** on every hit, DoT, reaction, and Duet!
    - `SWORD_QI_MERIDIAN`: Instead of adding `+18` Armor Penetration (`pen`) in `STATS` / `C.strike` (as stated in its spec and tooltip!), line 277 multiplies post-armor `dmg` by `1.09 * 1.28 = +39.5%` even if the caster is unarmed or casting non-sword poison spells!
    - Meanwhile, the other 5 archetypes (`ABYSSAL_FROST_MIRROR`, `EVERGREEN_LONGEVITY_SPRING`, `ASTRAL_VOID_BREATHING`,

DEVOURING_NETHER_VORTEX, VAJRA_IMMORTAL_BODY_BELL) **never multiply their combat effect by `stMult` at all**, AND ABYSSAL_FROST_MIRROR is missing its promised attacker-chill proc and Frost Jade Shield, while EVERGREEN_LONGEVITY_SPRING is missing its -6% critDmg tradeoff!

2.5 Remediation Blueprint ([MANTRA-CMB-02])

- Pass an explicit { isDotTick: true } / note = 'DOT_TICK' flag from C.statusTick and tickDomain so reactive Mantra procs (NETHER_VORTEX_SIPHON, EVERGREEN_SPRING_HEAL, VAJRA_BELL_REFLECT, SWORD_QI_COUNTER, PRIMORDIAL_YANG_FURNACE backlash) trigger **only on direct combat strikes**, never on DoT/Domain ticks.
- Gate EVERGREEN_SPRING_HEAL with a per-caster ICD (a._evergreenHealUntil = W.t + 4, i.e., once per combat round!), and multiply both EVERGREEN_SPRING_HEAL and NETHER_VORTEX_SIPHON by C.healMultiplier(a, W.t).
- Require melee proximity (C.unitDist(a, d) <= 2.5) for VAJRA_BELL_REFLECT and PRIMORDIAL_YANG_FURNACE fire backlash, and count their reflected damage toward the unified per-hit reflection cap (<= 22% attacker MaxHP).
- Fix stMult math from broken dmg * (1 + bonus) * stMult (1.15 * 1.28 = +47.2%!) to **scaling the bonus delta only**: 1 + bonus * stMult (1 + 0.15 * 1.28 = +19.2%!), and apply stMult consistently across all 7 archetypes' bonus deltas. Move SWORD_QI_MERIDIAN's +18 Penetration into STATS (f.pen += 18 * stMult when wielding Sword/Blade or on XIAN).

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3. [MANTRA-FORM-03] Grand Sect Formation Bias in Intra-Sect grudge Duels & Missing Officer AI

3.1 Sect Grand Formation Unfairly Helps Side A Murder Side B in Intra-Sect grudge Duels!

- **Location**: js/combat.js:300–309
- **Exact Code**:

```javascript
const aIsSectSide = !!(a && !a.foe && (!f || !C.sideOf(f, a.id) || (C.sideOf(f, a.id) === 'A' && f.kind !== 'duel' && f.kind !== 'spar' && f.kind !== 'tourney')));
```

- **Logic Bug**:

1. Intra-sect vendetta duels spawned from `W.duelQueue` use **`f.kind === 'grudge'`** (`combat.js:1218`), and leadership challenges use **`f.kind === 'leadership'`**! Neither `'grudge'` nor `'leadership'` is excluded in lines 302–303!
2. In a `grudge` or `leadership` duel, **both `f.A[0]` (challenger) and `f.B[0]` (defender/Sect Master) are members of the same sect**! Because line 302 checks `C.sideOf(f, a.id) === 'A'`, the challenger in `f.A[0]` receives **`+22%` damage (`SWORD_NET_SLAUGHTER`) or `-22%` damage taken (`IMMORTAL_AEGIS_DOME`) from the Sect's Grand Protective Array against their own Sect Master (`f.B[0]`)**!!
3. Similarly, `f.kind === 'wild'` (hunting beasts outside the sect walls) is not excluded, granting Sect Formation buffs outside the sect perimeter!

- **Fix**: Apply `SECT.FORMATION_WARFARE` combat bonuses **strictly when `f && f.kind === 'raid'`** (defending the sect against external invaders on Side `'A'`). And wire the `arraymaster` / `general` / `master` AI to automatically switch `activeStance` from `'FIVE_ELEMENT_NURTURE'` to `'IMMORTAL_AEGIS_DOME'` / `'SWORD_NET_SLAUGHTER'` during raids and revert to `'FIVE_ELEMENT_NURTURE'` once `W.threat.raid` ends!

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4. `[MANTRA-LIFE-04]` Fatal Old-Age Instant-Death Trap in `CULT.bestMantra(n)` & Element-Clash Blindness

4.1 Studying a Higher-Tier Non-Longevity Mantra Can Instantly Kill an Elderly Elder!

- **Locations**: `js/cultivation.js` (`CULT.bestMantra`, `CULT.lifeMax`), `js/ai.js:573` (`A.study.done`)
- **Chain-Reaction Proof**:
  1. `EVERGREEN_LONGEVITY_SPRING` Mantras (such as `MANTRA_JADE_SPRING` Tier 2 or `MANTRA_FIVE_ELEMENT_PROSPERITY` Tier 5) grant **`+25%` to `+28%` bonus `lifeMax`** while equipped in `n.cult.mantra`.
  2. Suppose a venerable Elder is `265` years old with base Stage-4 `lifeMax = 240`, kept alive and healthy (`effective lifeMax = 307`) by their Tier-5 `EVERGREEN_LONGEVITY_SPRING` Mantra.
  3. The Elder studies a Tier-6 offensive Mantra (`MANTRA_HEAVENLY_SWORD_QI` — `SWORD_QI_MERIDIAN`, `+0%` longevity).
  4. Upon finishing `A.study.done`, `CULT.bestMantra(n)` ranks Mantras purely by `tier` / `power`, automatically overwriting `n.cult.mantra = 'MANTRA_HEAVENLY_SWORD_QI'`.

5. **Instant Fatal Drop**: `CULT.lifeMax(n)` immediately plunges from `307` back to `240` (`age 265 > lifeMax 240`)! On the very next daily tick, the Elder suffers catastrophic old-age mortality and dies—**killed by reading a book in the library**!

- **Second Blindspot in `CULT.bestMantra(n)` — Element Clash Ignorance**:
  - A cultivator with a pure `WATER_ICE` Spirit Root who owns both a Tier-4 `WATER_ICE` Mantra (`RESONANCE`: `×1.28` cult rate, `-35%` deviation risk) and a looted Tier-5 `FIRE` Mantra (`CLASH`: `×0.60` cult rate, `+18%` Heart Demon/Deviation risk!) should **never** blindly equip the clashing Fire Mantra (`145 * 0.60 = 87` effective output vs `92 * 1.28 = 117.8`, plus lethal deviation risk!).
- **Remediation Blueprint**:
  - Upgrade `CULT.bestMantra(n)` to score candidate Mantras (filtering strictly to compatible path: `m.path === c.path || m.path === 'ANY' || (m.compatiblePaths && m.compatiblePaths.includes(c.path))`) using a holistic utility formula:
    1. **Lifespan Safety Hard Gate (CRITICAL)**: If `n.h.ageYears >= CULT.lifeMaxWithoutMantra(n) * 0.88`, apply an overwhelming survival weight (`+5000`) to `EVERGREEN_LONGEVITY_SPRING` / lifespan-extending Mantras so aging Elders *cherish* longevity scriptures rather than committing suicide!
    2. **Element Fit Multiplier**: Multiply raw `power / tier` score by `ELEM.evaluateTechElementFit(n, m).cultivationRateMult`.
    3. **Role & Qinggong Synergy**: Give `+18%` score to `SWORD_QI_MERIDIAN` for Sword Flight / Sword subpaths, `VAJRA_IMMORTAL_BODY_BELL` for Tank/Bodyguard roles, `ABYSSAL_FROST_MIRROR` when `heartDemon >= 35`, and `DEVOURING_NETHER_VORTEX` for `DEMON` / high-aggression cultivators.
  - Also call `n.cult.mantra = (CULT.bestMantra(n) || {}).id || null` inside `retraining.js` (`M.retrain`) so retrained disciples never keep an illegal Mantra from their old path!

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5. [MANTRA-AUTO-05] 100% Frozen Qi Stances, Unaware Meditation Seating, & Zero Meditation Mastery

5.1 NPCs Never Switch Qi Circulation Stances (`setMantraStance` Has Zero AI Callers!)

- **Verified Fact**: Searching the entire codebase for `setMantraStance` returns **only `cultivation.js`, `ui.js`, and `v550` test suite**. Every NPC in a natural simulation stays locked in `'HARMONIOUS_CIRCULATION'` forever!
- **Natural Autonomous Stance Logic to Add (`CULT.MANTRA_SYSTEM.evaluateAutonomousStance(W, n)`)**: Evaluate at `A.cultivate.start`, `SIM.hourly`, and `COMBAT.enter`:
  - **Switch to `'SERENE_PURIFICATION'` (🪷 เดินปราณสงบกลั่นบริสุทธิ์)** when:
    - `(n.cult.toxic || 0) >= 22` OR `(n.h.state.heartDemon || 0) >= 28` OR `(n.h.state.stress || 0) >= 68`, OR
    - `n` holds `VOW_ABSTAIN_PILLS` / `VOW_NO_MEDICINE` (using pure inner breathing to cleanse toxins instead of pills!), OR
    - `caution >= 72 && ambition < 50` while preparing safely for an upcoming breakthrough (`BN`).
  - **Switch to `'OVERDRIVE_COMBUSTION'` (🔥 เผาผลาญปราณเร่งตบะสุดขั้ว)** when:
    - `(n.h.state.energy || 0) >= 45` AND `(n.cult.toxic || 0) < 30` AND `(n.h.state.heartDemon || 0) < 35`, AND:
      - `ambition >= 74 && boldness >= 62`, OR `c.path === 'DEMON'` / `'VOID_HERETIC_PACT'`, OR
      - `n` has an active rivalry/tournament within 5 days or `W.threat.alert` is active!
    - **Safety Cutout**: Automatically drop out of `'OVERDRIVE_COMBUSTION'` when `energy < 22` (preventing `ASTRAL_VOID_BREATHING` walkers from collapsing to `0` energy mid-walk!), `toxic >= 48`, or `heartDemon >= 52` (unless crazed `heartDemon >= 80`, creating authentic *Qi Deviation* / ธาตุไฟเข้าแทรก stories!).
  - **Otherwise default to `'HARMONIOUS_CIRCULATION'` (☯️ โคจรปราณผสานฟ้าดิน)**.

5.2 Spatial Dharma Halos: No Seat-Seeking AI & No Orthodox-vs-Demonic Qi Interference

- **Locations**: `js/cultivation.js` (`stepMeditationResonance`), `js/ai.js` (`A.cultivate.where` / `AI.freeStation`)
- **Contextual Gaps**:

1. **Blind Seat Selection**: When an NPC goes to meditate (`A.cultivate`), `AI.freeStation` picks a cushion without checking who is meditating nearby. Junior disciples should prefer cushions within `6` tiles of a friendly Elder/Mentor radiating `PRIMORDIAL_YANG_FURNACE` (`+12%` cult speed), `ABYSSAL_FROST_MIRROR` (Heart Demon calming), or `EVERGREEN_LONGEVITY_SPRING` (healing)!
2. **Missing Demonic/Clash Interference**: Right now, a Demonic cultivator running `DEVOURING_NETHER_VORTEX` in `OVERDRIVE_COMBUSTION` right next to a `BUDDHA_DHARMA` or `XIAN` novice causes **zero Qi disturbance**! Adding mild `+stress / +heartDemon` friction when `DEVOURING_NETHER_VORTEX` (Overdrive) meditates within `4` tiles of orthodox disciples—while making `BUDDHA_DHARMA` / `ABYSSAL_FROST_MIRROR` suppress nearby deviation—creates emergent Peak segregation where demonic cultivators naturally retreat to secluded caves/peaks!
3. **Walking Cultivation (`ASTRAL_VOID_BREATHING`) Guardrails**: Pause walking `cult.prog` gain when `n.fled`, `n.h.state.hunger >= 88`, or `n.cult.prog >= CULT.need(n)` (at realm bottleneck).

5.3 Meditating (`A.cultivate`) Never Increases Mantra Mastery (`c.tm[c.mantra]`)!

- **Locations**: `js/ai.js` (`A.cultivate.tick`), `js/cultivation.js` (`CULT.rate`), `js/skills.js` (`SK.unlockedStage`)
- **Lore & Logic Bug**:
  - A Mantra (心法) is mastered by **meditating**, yet `c.tm` only increases when casting active skills in `COMBAT.useSkill`! Consequently, a secluded Elder who meditates on *Yi Jin Jing (`LEG_MANTRA_YIJIN_JING`)* or *Tai Xuan Jing (`LEG_MANTRA_TAIXUAN_JING`)* for 40 years in peaceful seclusion stays at **`tm = 0` (Stage 1/3)**!
  - And for non-legendary Mantras, `c.tm[c.mantra]` has `0%` effect on Mantra efficiency!
- **Fix**:
  - Every `30` ticks of `A.cultivate` (and `1.5x` faster in Closed-Door Seclusion or under `spirit_aurora`), call `SK.gainMastery(W, n, n.cult.mantra, 0.28 * (1 + comprehension / 150))`.
  - Feed Mantra mastery `tm = clamp(c.tm[c.mantra] || 0, 0, 100)` into `CULT.rate` and `MANTRA_SYSTEM` (`1 + (tm / 100) * 0.20`, up to `+20%` mastered circulation efficiency at 100 mastery), making decades of devotion to a signature Mantra genuinely rewarding!

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6. Implementation & Regression Checklist for GPT / Gemini (`03C`)

- [ ] **Catalog & Stat Parity**: Unify `CULT.allMantras` lookup across `JD.TECH.mantras` and `*_MANTRA` techniques without breaking `v550` schema checks; add/normalize a Tier-1 `GOD` starter Mantra; route Mantra combat stats in `stats.js` (`compute` & `getBaseFast`) by path `statWeights` instead of defaulting 31 Grand Dao paths to `+matk`; convert `SHADOW_SAMURAI` Mantra Qi effects into Bushido Focus/Stamina/Crit bonuses.
- [ ] **Combat Dharma Fixes (`combat.js:263–309`)**: Lazily initialize `mantraState`; block Mantra on-hit/on-struck procs from firing on `statusTick` DoTs and `tickDomain`; add 4-tick round ICD + `healMultiplier` check to `EVERGREEN_SPRING_HEAL`; enforce melee range (`<= 2.5`) and unified reflection cap on `VAJRA_BELL_REFLECT` / `PRIMORDIAL_YANG_FURNACE`; fix `stMult` to scale bonus delta `(1 + bonus * stMult)` across all 7 archetypes; restrict `FORMATION_WARFARE` combat buffs strictly to `f.kind === 'raid'`.
- [ ] **Lifespan & Retraining Safety**: Protect aging NPCs in `CULT.bestMantra(n)` from unequipping longevity Mantras (`EVERGREEN_LONGEVITY_SPRING`), weight by `ELEM.evaluateTechElementFit`, and re-resolve `c.mantra` inside `M.retrain`.
- [ ] **Emergent Autonomy & Meditation Mastery**: Implement `CULT.MANTRA_SYSTEM.evaluateAutonomousStance(W, n)`, aura-aware meditation station scoring in `A.cultivate`, Sect Officer `FORMATION_WARFARE` switching on raids, and Mantra `c.tm` mastery gain during `A.cultivate`.